## 1. ELIMINATE $I_2$

Eq. 1 $\times$ 7 : $35I_1 - 14I_2 = 70$

Eq. 2 $\times$ 2 : $-4I_1 + 14I_2 = 12$

$$31I_1 = 82$$

$$\boxed{I_1 = 2.65\,\text{mA}}$$

## 2. BACK-SUBSTITUTE

$$-2(2.645\ldots) + 7I_2 = 6$$

$$\boxed{I_2 = 1.61\,\text{mA}}$$

## 3. FIND $I_3$

$$I_3 = I_1 - I_2$$

$$\boxed{I_3 = 1.03\,\text{mA}}$$

**CHECKS:** KCL: $2.65 \approx 1.61 + 1.03$ KVL: both loop sums are $0\,\text{V}$